

Mark Dodici

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RESEARCH INTERESTS

Binary stars in dense clusters; transient phenomena in galactic nuclei; hierarchical triples and few-body dynamics; stellar tides; intermediate-mass black holes; celestial mechanics; galactic dynamics

EDUCATION

University of Toronto 2022 – (expected 2027)

Ph.D. candidate, Department of Astronomy & Astrophysics

Graduate researcher, Canadian Institute for Theoretical Astrophysics (CITA)

Visiting Graduate Student, Institute for Advanced Study (Winter 2026)

Advisors: Professors Yanqin Wu & Scott Tremaine

Princeton University 2018 – 2022

A.B. Astrophysical Sciences, *magna cum laude*. Certificate in Planets and Life

Thesis advisors: Dr. Christopher Spalding & Professor Jeremy Goodman

PUBLICATIONS 4 peer-reviewed (2 first-author); 61 citations (28 first-author)

FIRST AUTHOR

M. Dodici, 2026. Binary disruptions driven by massive disks around massive black holes. [arXiv:2607.26205](https://arxiv.org/abs/2607.26205).

M. Dodici, S. Tremaine, & Y. Wu, 2026. Dynamical evolution of stellar binaries in galactic centers. [ApJ](#), 1000, 226.

M. Dodici & S. Tremaine, 2024. Studying binary formation under dynamical friction using Hill's Problem. [ApJ](#), 972, 193.

CONTRIBUTING

B. Lewis et al. (incl. **M. Dodici**), 2025. Improving Undergraduate Astronomy Students' Skills with Research Literature via Accessible Summaries: A Case Study with Astrobites-based Lesson Plans. [Phys. Rev. Physics Education Research](#), 21, 010124.

B. Hensley, C. Murray, & **M. Dodici**, 2022. Polycyclic Aromatic Hydrocarbons, Anomalous Microwave Emission, and their Connection to the Cold Neutral Medium. [ApJ](#), 929, 23.

TALKS 19 total; 3 invited (*), 1 award-based (†)

AT CONFERENCES

* Galactic Nuclei Astrophysical Phenomena Sexten, Italy. Jul 2026

† AAS Division on Dynamical Astronomy 57 Chicago, IL. Jun 2026

CITA @ 40 / Kingston Meeting 2026 Toronto, ON. May 2026

Traversing the Galactic Center in Space and Time / IAU Symposium 405 Brno, Czechia. May 2026

MODEST-25 / IAU Symposium 398 Seoul, South Korea. Jun 2025

AAS Division on Dynamical Astronomy 56 Atlanta, GA. May 2025

AAS Division on Dynamical Astronomy 55 Toronto, ON. May 2024

AAS Division for Planetary Science 55 San Antonio, TX. Oct 2023

CITA Planet Day Toronto, ON. Aug 2023

Emerging Researchers in Exoplanet Science Symposium VIII New Haven, CT. Jun 2023

Great Lakes Exoplanets Area Meeting
AAS 240

Columbus, OH. Nov 2022
Pasadena, CA. Jun 2022

AT INSTITUTIONS

Galactic Center Group, Max Planck Institute for Extraterrestrial Physics	Garching, Germany. May 2026
Rossi Group, Leiden Observatory	Leiden, Netherlands. May 2026
* Dynamics Blackboard Talk, Institute for Advanced Study	Princeton, NJ. Mar 2026
* Astrophysics Seminar, Institute for Advanced Study	Princeton, NJ. Feb 2026
Compact Objects Group, CITA	Toronto, ON. Feb 2024
Lunch Talk, University of Toronto	Toronto, ON. Jan 2024, Apr 2026

FELLOWSHIPS, GRANTS, and AWARDS

Jui Lin (Allen) Yen Award	2026
Lachlan Gilchrist Fellowship (UofT School of Graduate Studies)	2026
Duncombe Student Research Prize (AAS Division on Dynamical Astronomy)	2026
Carl Reinhardt Fellowship in Astronomy	2025
Dunlap Institute Seed Funding Grant, <i>for Coding the Cosmos (outreach workshop series)</i>	2024
<i>for Age of the Universe (outreach workshop)</i>	2023
CITA Graduate Scholarship	2023
Hartmann Student Travel Grant (AAS Division for Planetary Science)	2023
C.a. Chant Fellowship in Astronomy	2022, 2023
David A. Dunlap Entrance Award	2022

LEADERSHIP and SERVICE ROLES

Referee, <i>ApJL</i> , <i>OJA</i>	2025 –
Coding the Cosmos (HS workshop series), <i>Co-director (2024 –)</i>	2023 –
Graduate Astronomy Students Association	2022 –
<i>Co-President (2024 – 2025)</i>	
<i>Courses & Qualifying Exams Committee (2022 – 2024, 2025 –)</i>	
Department Jamboree, <i>Organizer & Emcee</i>	2023 – 2025
UofT AstroTours (Public lecture/telescope events), <i>Co-director (2023 – 2024)</i>	2022 – 2024
Age of the Universe (HS workshop), <i>Co-organizer & Speaker</i>	2023

TEACHING ASSISTANT EXPERIENCE

The Sun and its Neighbours (AST101, Toronto)	Fall 2022, 2023; Summer 2026
<i>Lecturer, Summer 2026</i>	
Stars and Planets (AST221, Toronto)	Fall 2024, 2025
Introduction to Astrophysics (AST320, Toronto)	Winter 2025
Stars and Galaxies (AST201, Toronto)	Winter 2023, 2024
The Universe (AST203, Princeton)	Winter 2021, 2022

OTHER OUTREACH

UofT Astrotours, <i>Speaker</i>	2026
Pursue STEM (HS science program), <i>Workshop leader</i>	2025
Astrobites (Research blog), <i>Author & Education Study co-author</i>	2023 – 2025
Eclipse 2024 at the Toronto Public Library (Public lecture series), <i>Speaker</i>	2024
Solar Eclipse 2024: Indigenous Knowledge (Video), <i>Editor</i>	2024
Cosmos from your Couch (Astronomy video series), <i>Editor</i>	2023
ComSciCon Canada (Science Communication Workshop), <i>Attendee</i>	2023
Princeton Research Day (Virtual talk competition), <i>Award-winning presenter</i>	2021